

1(a)	all three correct = 1 mark ;	1														
	<table border="1"> <tr> <th rowspan="2">type of respiratory substrate</th><th colspan="2">energy value / kJ g⁻¹</th></tr> <tr> <th>approximately 17</th><th>approximately 37</th></tr> <tr> <td>carbohydrate</td><td>✓</td><td></td></tr> <tr> <td>lipid</td><td></td><td>✓</td></tr> <tr> <td>protein</td><td>✓</td><td></td></tr> </table>	type of respiratory substrate	energy value / kJ g ⁻¹		approximately 17	approximately 37	carbohydrate	✓		lipid		✓	protein	✓		
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	approximately 17	approximately 37														
carbohydrate	✓															
lipid		✓														
protein	✓															
1(b)(i)	respirometer ;	1														
1(b)(ii)	to, absorb / remove, carbon dioxide ;	1														
1(c)(i)	3 ;	1														
1(c)(ii)	1.33 ; ecf from (b)(i)	1														
1(d)(i)	any three from: 1 RQ value falls ; 2 data quote ; RQ = 0.9 at, start of torpor / 8:00 or maximum RQ = 0.98 and RQ = 0.75 at, end of torpor / 17:12 3 respire / metabolises, lipid / protein ; 4 AVP ; e.g. running out of available, carbohydrate / glycogen	3														

Question	Answer	Marks
1(d)(ii)	any two from: 1 during, winter / cold weather, when food is limited / to conserve energy / to reduce heat loss / AW ; 2 during, summer / hot weather / dry conditions, to, conserve water / prevent overheating ; 3 to avoid predation ; 4 AVP ; e.g. factor plus reason	2

Question	Answer	Marks
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